This study demonstrated the benefits of AECC on blastocyst quality compared with conventional embryo culture medium, in a broader category of patients referred to ART as opposed to other studies that concentrated on specific causes of infertility only. However, limitations of the study design should be taken into consideration; the analysis was performed using embryos rather than patients and a follow-up of children born following the treatments could not be conducted.

Computer Assisted Sperm Analysis (CASA)

There is insufficient evidence to permit conclusions regarding the use of this sperm function test. Study results to date have demonstrated low specificity, low sensitivity and a high rate of false positives.

The evaluation of sperm characteristics is part of the diagnostic process for infertile couples. Computer-aided semen analysis (CASA) uses a hardware- and software-based system to determine characteristics of sperm, including the kinematics/motility, concentration, and morphology.

In a recent systematic review conducted by Finelli et. al. (2021), the validity and reliability of computer-aided semen analyzers in performing semen analysis was examined. A total of 14 studies were included. Our results showed a high degree of correlation for sperm concentration and motility when analysis was performed either manually or by using a CASA system. However, CASA results showed increased variability in low (<15 million/mL) and high (>60 million/mL) concentration specimens, while sperm motility assessment was inaccurate in samples with higher concentration or in the presence of non-sperm cells and debris. Morphology results showed the highest level of difference, due to the high amount of heterogeneity seen between the shapes of the spermatozoa either in one sample or across multiple samples from the same subject. Overall, the study suggests CASA systems as a valid alternative for the evaluation of semen parameters in clinical practice, especially for sperm concentration and motility. However, further technological improvements are required before these devices can one day completely replace the human operator. Artificial intelligence-based CASA devices promise to offer higher efficiency of the analysis and improve the reliability of results.

EmbryoGlue

There is insufficient evidence supporting the clinical utility of EmbryoGlue. Further studies are needed to support improved clinical outcomes measures.

EmbryoGlue is a hyaluronan-enriched embryo transfer (ET) medium which aids in implantation of embryos, hence, improves pregnancy rates in in vitro fertilization-ET cycles (IVF-ET).

Singh, et. al. (2015) conducted a prospective case-control study at an assisted reproductive center of a tertiary care hospital. Forty-two women undergoing IVF, embryos were transferred into 50 µL of EmbryoGlue for 10 min prior to transfer inside uterine cavity. In the control group (n = 42), embryos were transferred to conventional blastocyst culture medium. Clinical pregnancy rate in the study group was 7% higher than in the control group. The difference, however, was not statistically significant. In